MEET DHAMECHA

Python Full-Stack & AI/LLM Engineer

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14 Projects **2+** Years Exp **80%** LLM Cost ↓ **100×** Speedup **30K+** Lines of Code

PROFESSIONAL SUMMARY

Results-driven Python Backend & AI Systems Developer with 2+ years of hands-on experience building scalable REST APIs, intelligent automation pipelines, and production-grade LLM-powered applications. Proficient in Django, FastAPI, and full-stack data workflows using Pandas and NumPy. Expert in modern AI engineering with LangChain, LangGraph, Hugging Face Transformers, OpenRouter, Ollama, and MCP (Model Context Protocol), with deep specialization in designing Retrieval-Augmented Generation (RAG) and Cache-Augmented Generation (CAG) systems, autonomous AI agent frameworks, and document intelligence platforms backed by Qdrant, ChromaDB, and PGVector. Proven ability to integrate multiple LLM providers (Gemini, Claude, OpenRouter, HuggingFace) with automatic failover and cost optimization. Comfortable shipping containerized microservices with Docker and Kubernetes, building CI/CD pipelines, and deploying to AWS infrastructure.

EXPERIENCE

TeamOpine Solutions — Surat, Gujarat **Jan 2025 – Present**

Backend Developer

Python · FastAPI · LangChain · LangGraph · RAG · CAG · MCP · Celery · Redis · PostgreSQL · Docker · Gemini API · Claude API · HuggingFace · Qdrant · ChromaDB · Selenium · Pandas

- Developed high-performance FastAPI microservices with role-based authentication, JWT security, Pydantic validation, and Celery + Redis background task processing — reducing endpoint response times by over 60%.
- Designed and deployed a multi-document RAG system supporting PDF, DOCX, XLSX, CSV, and image inputs with hybrid vector search (dense + sparse) using Qdrant and ChromaDB. Implemented page-level indexing (PageIndex) — tracking page numbers, section headers, and document metadata per chunk for precise source attribution and page-aware retrieval.
- Architected a production-grade RAG + CAG chatbot platform with real-time WebSocket communication, Redis pub/sub, and dual-layer semantic caching achieving sub-100ms cached retrieval (5–10ms) and sub-500ms full pipeline latency. Built vectorless PageIndex fallback using keyword-based BM25 retrieval with page-level granularity for scenarios requiring deterministic, non-embedding retrieval.
- Built MCP (Model Context Protocol) servers using FastMCP framework integrating Gemini 2.5-Flash and Claude API with multi-LLM AI agent frameworks unifying OpenRouter, HuggingFace, Gemini, and Claude.
- Architected a graph-based multi-agent automation framework using LangGraph and LangChain connected to WhatsApp and Telegram bots — achieving 80% reduction in LLM API spend via local Ollama models.
- Integrated Gemini 2.0 Flash Vision for an AI food nutrition prediction backend serving 1,000+ daily requests at sub-800ms average response time; collaborated on Power BI dashboards.
- Designed and deployed advanced web scraping pipelines using Selenium with headless Chrome (anti-detection strategies, proxy rotation, CAPTCHA-handling) targeting 12+ data portals — maintaining 95%+ uptime.
- Automated data transformation and export workflows using Pandas with Celery-powered task queues for fully automated data collection and delivery pipelines without manual intervention.

Zerobit Infotech — Surat, Gujarat **Jul 2024 – Dec 2024**

Software Engineer

Python · Django · Django REST Framework · FastAPI · Selenium · BeautifulSoup · PostgreSQL · MySQL · Ubuntu Server · Nginx · Git

- Built scalable Django and DRF web applications with JWT authentication, DTO-based handling, schema validation, and optimized PostgreSQL/MySQL integration for multi-tenant workflows.
- Developed FastAPI-based REST APIs alongside Django projects — implementing async endpoints, Pydantic data validation, and modular service-layer architecture for clean, maintainable backends.
- Built targeted web scrapers using Selenium (headless Chrome) and BeautifulSoup to extract and clean structured data from multiple US-based portals, automating previously manual data gathering tasks.
- Deployed and managed production applications on Ubuntu servers with Nginx reverse proxy, Gunicorn WSGI, and PostgreSQL connection pooling.

TECHNICAL SKILLS

Languages & Backend: Python · FastAPI · Django · DRF · Flask · Pydantic · SQLAlchemy · REST API

AI / LLM Engineering: LangChain · LangGraph · MCP / FastMCP · RAG / CAG · AI Agents · Tool Calling · Prompt Engineering · HuggingFace · Ollama · Tiktoken

RAG & Vector Databases: RAG Pipelines · CAG · PageIndex · Vectorless RAG · BM25 · Qdrant · ChromaDB · PGVector · Embeddings · Semantic Search · HNSW · Source Attribution

LLM Providers: Gemini 2.0/2.5 · Anthropic Claude · Llama 3 · Mistral · DeepSeek-V3.2 · OpenRouter

Data Science & ML: Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · Plotly

Scraping & Automation: Selenium · BeautifulSoup · XPath/CSS · n8n · Headless Chrome

Databases: PostgreSQL · MySQL · SQLite · Redis

DevOps & Cloud: Docker · Kubernetes · AWS EC2/S3/RDS · GitHub Actions · GitLab CI · Nginx · Gunicorn

BI, Tools & Practices: Power BI · DAX · Chart.js · D3.js · Git · Celery · WebSocket · Postman · Pytest

EDUCATION

B.Sc. Information Technology

2020 – 2024

J.P. Dawer Institute — Veer Narmad South Gujarat University, Surat

KEY PROJECTS

01. MCP RAG + CAG Chatbot Platform

FastAPI · LangChain · Qdrant · Redis · Celery · WebSocket · Docker · Gemini · Claude

Production-grade RAG + CAG chatbot with real-time WebSocket, 5-microservice Docker Compose, dual-layer semantic caching (5–10ms hits), MCP server integrating Gemini 2.5-Flash + Claude API. Implements PageIndex for page-level document indexing with vectorless BM25 fallback retrieval.

02. Database Insight Assistant

Django DRF · Qdrant · Redis · Celery · Chart.js · D3.js · Docker · Gemini · Groq · HuggingFace · OpenRouter

Enterprise-grade local-first AI database analytics platform. Django REST API with 60+ endpoints across 77 view classes, 30K+ LOC, 5-service Docker Compose (Django, Redis, Qdrant, Celery, Flower). 4-agent orchestration (SQL, Analyst, Reporter, Monitor) with automatic LLM failover across 4 providers. Power BI-like dashboard engine with 40+ DAX-like measures (YoY, MoM, YTD, running totals, percentiles), AI single-prompt dashboard generation, drag-and-drop widget builder, and portable .dbia export format. Glassmorphism frontend with Chart.js 4.x, D3.js force-directed schema graphs. Redis 256MB LRU cache delivering 100x speedup. Supports SQLite, PostgreSQL, MySQL, MSSQL with NL-to-SQL generation, statistical analysis, data quality monitoring, and vector semantic search via Qdrant + Sentence Transformers.

03. AI Agents with MCP Framework

MCP · FastMCP · LangChain · OpenRouter · Gemini · HuggingFace · Rich CLI

Multi-module AI agent framework demonstrating LLM vs Agent paradigm. FastMCP server with 4 tools, DatabaseAIAgent with multi-database NL-to-SQL, streaming responses.

04. Document Intelligence System

LangChain · Qdrant · ChromaDB · PGVector · FastAPI · Ollama · Tesseract OCR

Full document intelligence platform — upload PDF, DOCX, XLSX, images and converse with content via RAG pipeline. PageIndex-based page-level chunking with page number tracking, source attribution showing exact page references. Multi-tenant isolation with local LLM option.

05. Database Communicator RAG

Gemini API · ChromaDB · Sentence Transformers · SQLAlchemy · MySQL · Pandas · Python

Production-ready RAG pipeline converting plain-English questions into optimized SQL and returning answers in natural language. Multi-stage architecture: schema vector store (Chroma DB + Sentence Transformers) for context injection, Google Gemini SQL generation, result execution, NL answer synthesis. Semantic query cache with configurable similarity threshold (0.88 cosine) delivering ~5ms cache hits vs. full LLM round-trips, with statistics tracking and cache hit rate monitoring. Dynamic schema extraction populates vector store with table/column definitions enabling context-aware SQL across complex joins. 15+ core functions, 5 major RAG stages, 100% Python.

06. Multi-Agent AI Automation Platform

LangGraph · LangChain · Ollama · FastAPI · WhatsApp API · Telegram Bot · PostgreSQL

Graph-based multi-agent framework with 4 node agents connected to WhatsApp/Telegram bots. Users send commands and receive chart images directly in chat. 80% API cost reduction.

07. AI Food Nutrition Analyzer

FastAPI · Gemini 2.0 Flash Vision · AWS EC2 · RDS · Scikit-learn

Backend for Android app — food image analysis via Gemini Vision AI returning per-dish nutrition breakdowns. 1,000+ daily requests at sub-800ms response time.

08. ML Prediction & Analytics

Scikit-learn · Pandas · NumPy · SHAP · Matplotlib · Seaborn · FastAPI

End-to-end ML workflows: EDA, feature engineering, hyperparameter tuning with GridSearchCV, k-fold cross-validation, and SHAP interpretability for churn prediction & demand forecasting.

09. Power BI Analytics Dashboards

Power BI · DAX · Python · PostgreSQL · Pandas · NumPy · Direct Query

College attendance tracking & inventory management dashboards with drill-through views, live PostgreSQL Direct Query, DAX-calculated KPIs, and automated email alerts.

10. Secure Microservice API

FastAPI · Celery · Redis · SQLAlchemy · PostgreSQL · Pytest · GitLab CI

Production FastAPI microservice with OAuth2 + JWT, refresh token rotation, Celery + Redis async tasks, 85%+ test coverage, and dual-database support with zero-downtime migrations.

11. Full-Stack E-Commerce with Finance

Django · PostgreSQL · Pandas · OpenPyXL · Nginx · Gunicorn · Ubuntu

Production e-commerce app with GST invoicing, profit tracking, automated Excel reports. Database optimization achieved 4.3x API speedup (820ms to 190ms).

12. Multi-Site Scraping & Automation

Selenium · BeautifulSoup · Pandas · n8n · SQLite · Proxy Rotation · Stealth

Automated scraping framework targeting 12+ US portals. 500K+ records/week with anti-detection strategies, 95%+ uptime, and n8n workflow orchestration.

13. Amazon Scraper — ETL Pipeline

Selenium · Pandas · MySQL · WebDriverWait · Jupyter · CSV

End-to-end Amazon.in laptop scraper with multi-page pagination, stale element recovery, 64KB+ structured dataset extraction, and complete ETL to MySQL.

14. Pandas Data Visualization & EDA

Pandas · NumPy · Matplotlib · Seaborn · SciPy · Jupyter · Statistical Analysis

Comprehensive EDA workflow — groupby aggregations, pivot tables, multi-dimensional visualizations (heatmaps, pair plots, facet grids), and statistical analysis with actionable insights.

CORE STRENGTHS

▸ **Deep LLM Internals:** Tokenization, context windows, attention mechanisms & advanced prompt engineering for open-source and proprietary models.

▸ **MCP Architecture:** Model Context Protocol server architecture — tool design & AI agent-to-service integration using FastMCP framework.

▸ **Vector Search Systems:** Embedding model selection, hybrid retrieval strategies, chunk sizing & re-ranking for production RAG systems. Page-level indexing (PageIndex) with metadata tracking.

▸ **PageIndex & Vectorless RAG:** Page-level document indexing with page number tracking, section-aware chunking & source attribution. BM25 keyword-based fallback retrieval for deterministic, non-embedding search.

▸ **Production AI Safety:** Rate limiting, prompt injection prevention, SQL injection protection & secure API key management.

▸ **Full DevOps Lifecycle:** Multi-service Docker, CI/CD pipelines, AWS IAM & security configuration for containerized deployments.

WHAT I BRING

▸ **AI Engineering Depth:** Deep understanding of LLM internals — tokenization, context windows, attention mechanisms, and RAG/CAG architecture design. Specialized in PageIndex-based document retrieval and vectorless RAG strategies. Proven ability to integrate Gemini, Claude, Ollama, and HuggingFace models into production systems with cost optimization (80% API spend reduction achieved).

▸ **Backend & System Design:** Comfortable architecting complex distributed systems — async microservices, Redis pub/sub, Celery task queues, WebSocket real-time comms. Strong focus on performance optimization: 4.3x API speedup via query optimization, 100x speedup via intelligent caching layers.

▸ **Data & Analytics:** Full data pipeline ownership — collection (Selenium, 500K+ records/week), transformation (Pandas, ETL), storage (PostgreSQL, vector DBs), analysis (EDA, ML, SHAP), and visualization (Power BI, Plotly dashboards). Experienced replacing manual Excel workflows with automated pipelines.